Experiment No. 7
Case-study on the implementation of AI in Autonomous Vehicles.
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Aim: Case-study on the implementation of AI in Autonomous Vehicles.

Objective:

• To understand the role of Artificial Intelligence in autonomous vehicles.
• To study different AI techniques used in self-driving cars.
• To analyze real-world applications of AI in transportation systems.
• To explore challenges and future scope of autonomous vehicles.

Theory:

Autonomous Vehicles:

Autonomous vehicles (self-driving cars) are vehicles that can operate without human intervention using Artificial Intelligence, sensors, and advanced algorithms.

Role of AI in Autonomous Vehicles:

AI enables vehicles to:
Perceive surroundings
Make decisions
Navigate safely

Key Technologies Used:

Computer Vision: Used for object detection (cars, pedestrians, traffic signals).

Machine Learning / Deep Learning: Used for decision-making and prediction.

Sensor Fusion: Combines data from: Cameras, LiDAR, Radar

Path Planning Algorithms: Determines safe route and movement.

Control Systems: Controls steering, braking, and acceleration.

Working of Autonomous Vehicles:

- Sensors collect environmental data
- AI processes and interprets data

- System identifies objects and obstacles
- Decision-making algorithms determine actions
- Vehicle executes movement safely

Case Study: Tesla, Inc. Autonomous Driving System

One of the most prominent real-world implementations of AI in autonomous vehicles is developed by Tesla, Inc..

Tesla Autopilot System:

Tesla uses an advanced Autopilot system that enables semi-autonomous driving using AI.

Working of Tesla AI System:

- **Tesla vehicles are equipped with multiple cameras, ultrasonic sensors, and radar.**
- **The system uses computer vision and deep learning models to detect:**
  - **Vehicles**
  - **Pedestrians**
  - **Traffic lights and road signs**
- **AI models process real-time data to understand surroundings and predict risks.**
- **Neural networks are trained using large-scale data collected globally.**
- **Based on predictions, the system performs:**
  - **Lane keeping**
  - **Automatic braking**
  - **Adaptive cruise control**
- **The system improves continuously through over-the-air updates.**

Outcome:

The implementation of AI in Tesla vehicles results in:

- **Reduced driver workload**
- **Improved road safety**
- **Enhanced driving efficiency**
- **Semi-autonomous driving capabilities**

Challenges in Autonomous Vehicles:

- **Safety and reliability concerns**
- **High computational requirements**
- **Ethical decision-making (e.g., accident scenarios)**
- **Adverse weather conditions affecting sensors**
- **Legal and regulatory issues**

Future Scope:

- **Fully autonomous (Level 5) vehicles**
- **Smart traffic systems integration**
- **AI-powered transportation networks**
- **Reduced accidents and improved mobility**

Conclusion:

Artificial Intelligence plays a crucial role in autonomous vehicles by enabling perception, decision-making, and control. Systems like those developed by Tesla, Inc. demonstrate the real-world impact of AI in transportation. While challenges remain, AI-driven vehicles have the potential to revolutionize the future of mobility.